

An Exact Computer-Assisted Lower Bound for Packing 17 Unit Squares in a Square

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<https://github.com/Mira-acc/17squares>

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Abstract

Let $s(n)$ denote the least side length of a square that contains n pairwise interior-disjoint unit squares, with arbitrary orientations. We prove by an exact computer-assisted argument that

$$s(17) > \frac{4\,450\,837}{1\,000\,000} = 4.450837.$$

The previous lower bound recorded in Friedman’s survey was $(40\sqrt{2} + 19)/17 \approx 4.4452083821$. Our proof gives sixteen rational points in the square of side 4.450837 and certifies that every contained unit square, at every orientation, contains one of these points strictly in its interior. The result then follows by the pigeonhole principle. The universal geometric assertion is checked by a 3,341,159-node recursive bisection of the three-dimensional pose space. All decisive comparisons are exact integer inequalities. A fast C++ checker and separately implemented arbitrary-precision C++ and Python checkers validate the same certificate.

1 Introduction

For $L > 0$, write

$$Q_L = [0, L]^2.$$

Let $s(n)$ be the minimum L for which n pairwise interior-disjoint unit squares can be placed in Q_L , with arbitrary translations and rotations. The minimum is attained by the standard compactness argument: centers and orientations range over a compact parameter space, and the containment and nonoverlap constraints are closed. This classical packing problem has a long history; Friedman’s dynamic survey collects the principal constructions and lower-bound methods[1]. For $n = 17$, the survey records the lower bound

$$s(17) \geq \frac{40\sqrt{2} + 19}{17} = 4.445208382054341\dots,$$

attributed there to Trevor Green. The best known construction is due to John Bidwell and has side approximately 4.67553009360455[1, 3].

Our result is the following.

Theorem 1. *No packing of seventeen unit squares exists in Q_L for*

$$L = \frac{4\,450\,837}{1\,000\,000}.$$

Consequently,

$$s(17) > 4.450837.$$

Table 1: The sixteen rational witness points.

i	x_i	y_i	i	x_i	y_i
1	0.967095	0.988674	9	0.925919	2.658468
2	1.584591	0.989280	10	1.494140	2.660211
3	2.477806	0.929750	11	2.483891	2.663100
4	3.452882	0.920541	12	3.452714	2.667020
5	0.998081	1.794317	13	0.998038	3.524908
6	1.992852	1.799448	14	1.963307	3.513845
7	2.973306	1.794450	15	2.850741	3.459296
8	3.522479	1.795033	16	3.456188	3.456188

The improvement over the previously recorded lower bound is approximately 0.00562861795. The proof uses the unavoidable-point method associated with Stromquist and developed in this setting by Friedman[1, 2]. The new contribution is an explicit rational point set together with an exact, independently checkable certificate covering the full continuum of square positions and orientations.

2 Strictly unavoidable points

Definition 2. A finite set $P \subseteq Q_L$ is *strictly unavoidable for unit squares* if every unit square $U \subseteq Q_L$ satisfies

$$P \cap \text{int}(U) \neq \emptyset.$$

Lemma 3 (Pigeonhole lemma). *If Q_L contains a strictly unavoidable set of $n - 1$ points, then no packing of n pairwise interior-disjoint unit squares exists in Q_L .*

Proof. Suppose $U_1, \dots, U_n$ were such a packing. For each i , choose $p_i \in P \cap \text{int}(U_i)$. If $p_i = p_j$ for $i \neq j$, then that point lies in both square interiors, contradicting interior-disjointness. Thus the map $i \mapsto p_i$ is an injection from an n -element set into an $(n - 1)$ -element set, which is impossible. $\square$

The use of interiors is important. Boundary witnesses could be shared by several touching squares and would not imply the injection in Lemma 3.

For Theorem 1, the set P consists of the sixteen rational points in Table 1. Every displayed decimal is exact, with denominator 10^6 .

3 Parameterizing a unit square

A unit square is unchanged by a rotation through $\pi/2$. We may therefore represent every orientation by an angle

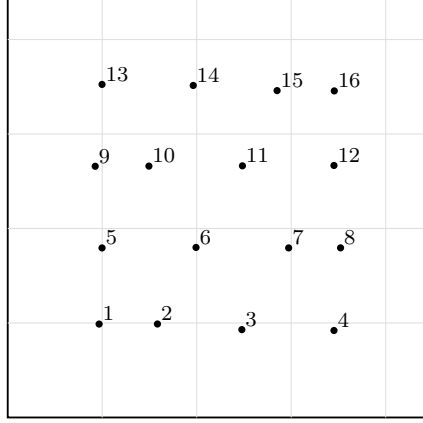
$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}, \quad t = \tan \theta \in [-1, 1].$$

Let (x, y) be the center of the square, and let $p = (p_x, p_y)$ be a fixed point. Put

$$\Delta x = p_x - x, \quad \Delta y = p_y - y, \quad r = \sqrt{1 + t^2}.$$

In coordinates aligned with the square, the displacement from the center to p is

$$\left(\frac{\Delta x + t\Delta y}{r}, \frac{-t\Delta x + \Delta y}{r} \right).$$



$$L = 4.450837$$

Figure 1: The witness set in the containing square. The arrangement is close to a distorted 4×4 grid, but the proof uses the exact rational coordinates in Table 1.

Consequently, p lies strictly inside the square exactly when

$$4(\Delta x + t\Delta y)^2 < 1 + t^2, \quad (1)$$

$$4(-t\Delta x + \Delta y)^2 < 1 + t^2. \quad (2)$$

The axis-parallel half-extent of a unit square with parameter t is

$$h(t) = \frac{1 + |t|}{2\sqrt{1 + t^2}}.$$

Thus containment in Q_L requires

$$h(t) \leq x, y \leq L - h(t).$$

The function $u \mapsto (1 + u)/(2\sqrt{1 + u^2})$ is nondecreasing on $[0, 1]$. This monotonicity supplies a simple exact test for pose boxes that cannot contain any feasible square.

4 The subdivision certificate

The pose space is parameterized by (x, y, t) . The checker begins with the rational box

$$\mathcal{B}_0 = \left[\frac{1}{2}, L - \frac{1}{2} \right]^2 \times [-1, 1],$$

which contains every feasible pose. It recursively bisects boxes in one of the three coordinates. Each node is encoded by one byte:

Byte	Meaning
0	the current box contains no feasible square pose;
1, ..., 16	the indicated point lies strictly inside every pose in the box;
17, 18, 19	bisect the box in x , y , or t , respectively.

The verifier distrusts every leaf claim and recomputes it from the box endpoints.

4.1 Infeasible leaves

For a box

$$\mathcal{B} = [x_0, x_1] \times [y_0, y_1] \times [t_0, t_1],$$

let

$$u = \min\{|t| : t \in [t_0, t_1]\}.$$

Since $h(t) \geq h(u)$ throughout the box, the box is infeasible if any one of

$$x_1 < h(u), \quad L - x_0 < h(u), \quad y_1 < h(u), \quad L - y_0 < h(u)$$

holds. The checker clears denominators and squares nonnegative quantities, so this test is performed using integers only.

4.2 Witness leaves

Fix a witness point p . Exact interval arithmetic over $\mathcal{B}$ gives intervals containing

$$A = \Delta x + t\Delta y, \quad B = -t\Delta x + \Delta y.$$

Write M_A and M_B for the maximum absolute endpoint of these intervals. Because $1 + t^2 \geq 1 + u^2$ throughout the box, the two strict inequalities

$$4M_A^2 < 1 + u^2, \quad 4M_B^2 < 1 + u^2$$

prove simultaneously that p is strictly inside every square represented by that box. Again, the implementation stores rational endpoints on fixed grids and clears every denominator before comparison.

The interval enclosure for the bilinear term $t\Delta y$ is obtained by taking the minimum and maximum of its four corner products; the same is done for $(-t)\Delta x$. No floating-point approximation enters a leaf decision.

4.3 Completeness of the tree

The checker traverses the tree from $\mathcal{B}_0$, applying every split exactly. It rejects an early end of file, an unknown opcode, a false leaf claim, or any trailing byte. Finally it checks the full-binary-tree identity

$$\#\text{leaves} = \#\text{split nodes} + 1.$$

It follows inductively that the accepted leaves form a cover of the complete root box. Every feasible pose is therefore covered by a witness leaf, proving that the sixteen points are strictly unavoidable.

Lemma 3 now rules out a packing in Q_L , and therefore in every smaller container as well. Thus $s(17) \geq L$. The minimum defining $s(17)$ is attained by compactness, so equality would give a packing in Q_L . Hence $s(17) > L$, proving Theorem 1.

5 Exact arithmetic and reproducibility

The certificate uses the scales

$$S_t = 2^{40}, \quad S_x = 10^6 2^{40}.$$

The center and slope endpoints are stored as integers on these fixed rational grids, while the witness points have denominator 10^6 . The accepted tree has

Total nodes	3,341,159
Split nodes	1,670,579
Witness leaves	1,355,082
Infeasible leaves	315,498
Maximum depth	81

The certificate consists of one byte per node. Its SHA-256 digest is

49ee8134ccf636224432d235fc3c1db2f8d6a567efe9e4c08b2e81639172cfef.

The repository provides three checkers:

1. a fast C++ checker using bounded wide integers and the same arithmetic header as the generator;
2. a separately implemented C++ checker using arbitrary-precision integers;
3. a separately implemented Python checker using arbitrary-precision integers.

The generator and its search heuristics are not part of the trusted proof core: each checker reads the certificate as untrusted input and recomputes every leaf claim. The two arbitrary-precision checkers do not reuse the generator’s arithmetic routines, and so do not rely on a fixed-width overflow analysis. From the repository root, the command

```
bash ./verify_all.sh
```

regenerates the deterministic certificate, checks its hash, and executes the three verifiers. The proof package is available at <https://github.com/Mira-acc/17squares>.

6 Discussion

This result is a small numerical improvement, but a complete theorem. The mathematical proof has two layers: the short unavoidable-point argument in Section 2 and an exact finite certificate for its universal geometric premise. The point search and the certificate-generation heuristic may be useful for finding the proof, but neither must be trusted to verify it.

The current gap remains substantial:

$$4.450837 < s(17) \leq 4.67553009360455 \dots$$

The certificate does not imply that the sixteen-point set is optimal, nor does it provide evidence that Bidwell’s construction is globally optimal. Natural next steps are external reproduction, formal verification of the small trusted kernel, optimization of unavoidable point sets, and construction searches informed by the least-covered poses.

Disclosure. The computation and initial exposition were developed with assistance from OpenAI’s GPT-5.6 Pro under human direction. The model is not listed as an author. The mathematical claim rests on the explicit point set, certificate, and executable exact checkers.

Status. This is a working preprint. At the time of writing, the certificate has been reproduced with the included exact checkers but has not undergone independent peer review.

References

- [1] E. Friedman, Packing unit squares in squares: A survey and new results, *Electron. J. Combin.*, Dynamic Survey DS7 (2009), first published 1998. <https://doi.org/10.37236/28>.
- [2] W. Stromquist, Packing 10 or 11 unit squares in a square, *Electron. J. Combin.* 10(1) (2003), R8. <https://doi.org/10.37236/1701>.
- [3] D. Ellsworth, Squares in squares, high-precision online packing catalogue, accessed 5 August 2026. https://kingbird.myphotos.cc/packing/squares_in_squares.html.